European Innovation Council

EIC Pathfinder Portfolio

PRECISION NUTRITION

Strategic plan

BRUSSELS, DECEMBER 2025



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ABBREVIATIONS

EFSA	European Food Safety Authority
EIC	European Innovation Council
EISMEA	European Innovation Council and SMEs Executive Agency
GDPR	General Data Protection Regulation
PM	Programme Manager
PO	Project Officer
TRL	Technology Readiness Level
WP	Work Package

1. INTRODUCTION

Background

Food is much more than the calories we eat. It is a source of building blocks for our bodies and the key to our well-being and longevity. On the other hand, inadequate diets can lead to developing various Food-related Health Conditions, like obesity and malnutrition and Noncommunicable Chronic Diseases such as Type 2 diabetes and hypertension.

Current dietary guidelines predominantly offer generalised recommendations aimed at achieving caloric and nutrient reference intakes, however, these guidelines often lack stratification based on critical factors such as age, gender, lifestyle and genetic predispositions.

Despite deepening knowledge in different scientific disciplines, further interdisciplinary research is needed to better understand the complex role of nutrition on human health. Precision nutrition is rapidly advancing with the help of cutting-edge technological innovations in different scientific domains, such as Artificial intelligence is expected to play a pivotal role in analysing complex biological data and integration of large Multiomics data sets and data from previous relevant research.

As precision nutrition continues to evolve, it holds the promise of transforming public health through novel food ingredients, nutritional supplements and tailored dietary guidance, while taking into consideration factors such as genetics, metabolism and lifestyle, and ultimately aiming to prevent chronic diseases and promote overall well-being.

The Challenge

In 2023, the European Innovation Council (EIC) launched the Pathfinder Challenge call "Precision Nutrition". The aim is to investigate the role of diet in Food-related Health Conditions and Noncommunicable Chronic Diseases by exploring causal relationships among diet, the gut microbiome, and glycans, with the potential to personalize human diet. The Challenge seeks to identify food ingredients, food technology processes, additives, and dietary patterns that positively or negatively affect human health and aging, and to develop recommendations for reformulating food products with fewer or no additives. By understanding these interactions, the selected projects should provide evidence for upgrading current dietary guidelines, ultimately reducing the incidence of Food-related Health Conditions and Noncommunicable Chronic Diseases and improving the quality of life for affected individuals.

Under this EIC Pathfinder Challenge call, eight projects were selected for funding forming the EIC Precision Nutrition Portfolio (referred to herein as "portfolio"). The portfolio projects

integrate nutrition, microbiome, and glycan studies to explore their interactions with immunometabolism and human health, focusing on novel and sustainable foods as insects, plant-based, processed, and fermented foods. They combine various research methods, including advanced ex-vivo models, observational studies, nutritional trials, food technology and machine learning approaches, while considering regulatory aspects and building on prior work by European Food Safety Authority (EFSA).

About the portfolio strategic plan

The portfolio strategic plan summarises the governance and structure, as well as the common activities planned within the portfolio, in accordance with the Challenge Guide. The plan serves as a common basis for the portfolio and was developed jointly between the consortia, the EIC Programme Manager (PM) and the EIC Project Officers (POs), Tables 1 and 2. It will be assessed internally once per year. The implementation of the Strategic Plan will be reported on through the publishing of Annual Portfolio Activity Reports.

CONTRIBUTORS	PROJECT
Morten Rasmussen, Rasmus Riemer Jakobsen, Robert Jan Brummer, Kenneth Barfod	BUGS4URATE
Maria Monsalve, João Carlos Serafin, Varela, Diana Martin, James Webb, Angela M Valverde	DIBAN
Anne Salonen, Anne-Maria Pajari, Robert Jan Brummer, Sari Bäck	FIBRE-MATCH
Jordi Merino, Eran Segal, Ellen Blaak, Hans Wandall, Emilie Weynants	GLUCOTYPES
Joeri Beauprez, Ut Van Nguyen, Joris Verster, Saskia Braber, Ingrid van den Bosche	INTERACTHOMIG
Ricardo Ramos, Celso Reis, Pedro Mena, Agustin Rodriguez, Cristina M Fernandez	MENTORING
Seppo Vainio, Gabriella Pocsfalvi, Terhi Ruuska-Loewald, Michael Hust, Susanna Kaisto	NUTRIEV
Lelja Imamovic, Tim Swartz, Nouredine Farah, Lysiane Jubin	NUTRIMMUNE

Table 1: Contributors from the projects in the portfolio.

CONTRIBUTORS	ROLE
Ivan STEFANIC	EIC Programme Manager for Food Chain Technologies, Novel & Sustainable Food
Christiane WILZECK	Project Officer, EIC Pathfinder Unit
Ioannis MANOLIDIS	Project Officer, EIC Pathfinder Unit
Johannes BÜNZ	Portfolio Officer, EIC Programme Managers Office
Danilo PIETRETTI	Portfolio Officer, EIC Programme Managers Office
Xenia CHRISTOFORIDOU	Trainee at the EIC Programme Managers Office
Aron DIENING	Trainee at the EIC Programme Managers Office

Table 2: Contributors from the European Innovation Council.

2. THE PORTFOLIO

Composition and working methods

The EIC Precision Nutrition Portfolio consists of the following 8 projects:

PROJECT NUMBER	ACRONYM	TITLE	PROJECT COORDINATOR	START DATE	END DATE
101161281	BUGS4URATE	Precision Nutrition Strategies for Investigating Probiotic, Diet, Glycan, and Microbiome Factors in Hyperuricemia and Gout Prevention	Morten Rasmussen	1/9/2024	31/8/2027
101161484	FIBRE-MATCH	Precision nutrition approach for increasing fibre intake and health using gut microbiota-matched sustainable fibre sources	Anne Salonen	1/9/2024	31/8/2028
101161110	INTERACTHOMIG	Understanding the potential of modulating Host-Microbiome-Glycan interactions ("the triangle of sweetness") to tackle non-communicable diseases	Joeri Beauprez	1/9/2024	31/8/2028
101162297	MENTORING	Addressing malnutrition and metabolic health in non-communicable	Ricardo Ramos	1/9/2024	31/8/2028

		diseases through precision Nutrition			
101161353	NUTRIEV	Development of nutritional vesicles for precision diagnostics and therapeutics for metabolic diseases	Seppo Vainio	1/9/2024	31/8/2028
101162457	NUTRIMMUNE	Precision Nutrition to optimize immune response for metabolic health	Karine Clément	1/9/2024	31/8/2028
101161509	GLUCOTYPES	Glucose variability patterns for precision nutrition in diabetes	Jordi Merino	1/10/2024	30/9/2028
101162517	DIBAN	Linking Intestinal Bacteria and Host Metabolism to Tackle Type 2 Diabetes with Novel Food	Maria Monsalve & Angela M Valverde	1/10/2024	30/9/2028

Table 3: Projects of the EIC Precision Nutrition Portfolio.

To establish a common framework for the portfolio, all projects have a separate Portfolio Work Package (WP) in their Grant Agreements. This WP covers the person-months for the portfolio activities. The portfolio kick-off meeting took place in February 2025 in Brussels and set the milestones for the portfolio objectives and suggested activities.

All projects are expected to contribute actively and continuously to the portfolio. The EIC PM works with the projects to define the governance of the portfolio, to establish expectations from the projects' collaboration, to define rules for resources and data sharing in accordance with the existing legal base,[1] and to identify the common exploitation, communication, and dissemination strategies for the portfolio.

[1] Specifically, Annex 6 of the EIC 2023 Work Programme.

Portfolio governance

The portfolio is governed on two levels, **strategic** and **operational** (see Figure 1), which are connected through the **Steering Committee**. This consists of the EIC Programme Manager, the EIC Project Officers, the project coordinators, and the respective chairs of the portfolio working groups. The committee meets online once every three months to report about the progresses in the portfolio working groups and discuss further steps.

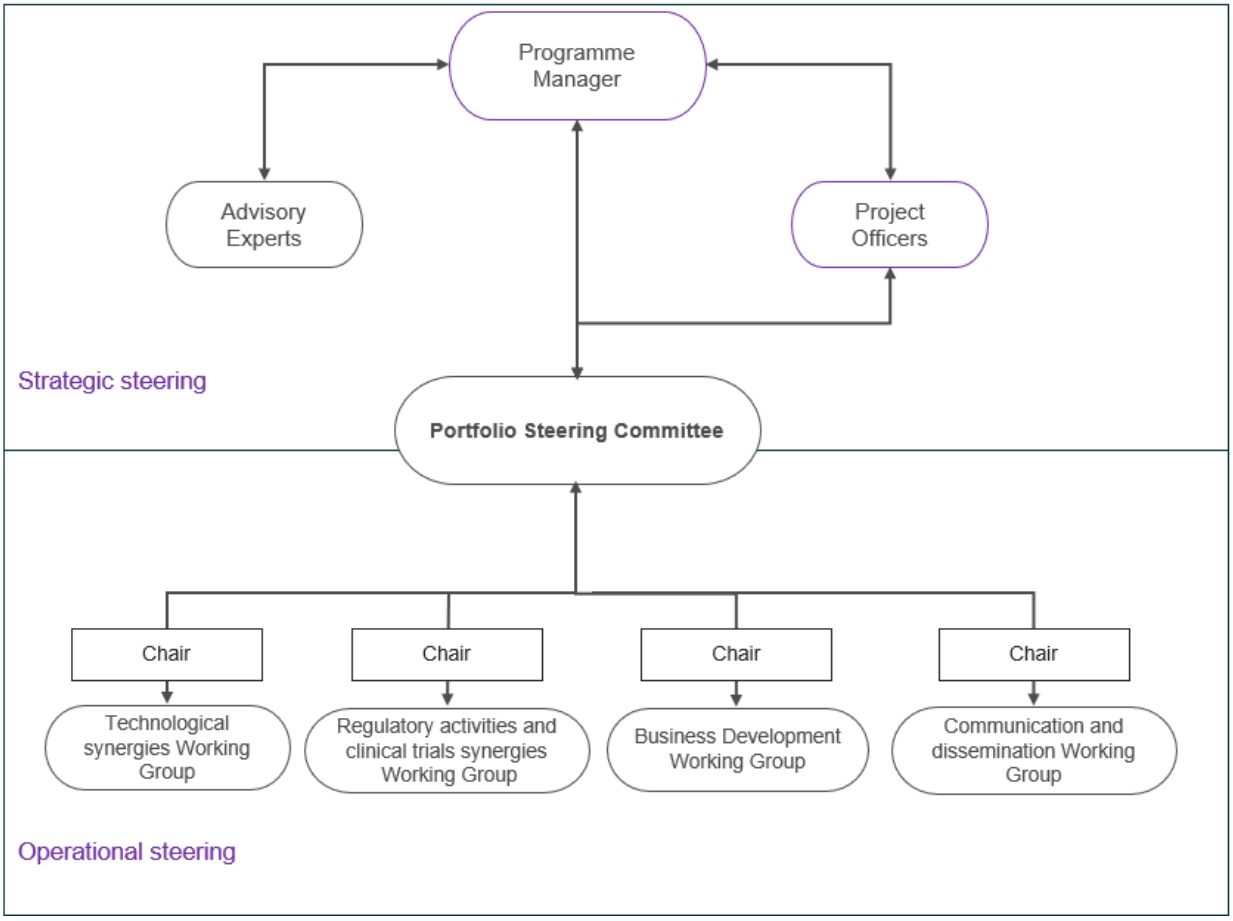


Figure 1: Governance structure and interactions among the key actors of the portfolio.

Strategic steering

The **Strategic Steering** of the portfolio is carried out by the EIC PM in close collaboration with the POs and the advisory external experts. The experts are selected by the PM based on their proven track record in the field and because of specific collaboration ideas together with and for the Working Groups of the portfolio. All experts sign a Non-disclosure Agreement (NDA) with EISMEA binding them to confidentiality. The experts should propose and conduct activities beneficial to the whole portfolio.

Operational steering

The operational steering is undertaken by the **portfolio Working Groups (WGs)**. Four WGs have been established:

1. Technological synergies,
2. Regulatory activities and clinical trials synergies,
3. Business development, and
4. Communication and dissemination.

All projects have nominated representatives for each of the four WGs to ensure targeted workflows and a clear hierarchy within the consortium. Each WG is led by a chair elected between the members of the group, or in the absence of volunteers, by temporary appointment. The chairs participate in the Steering Committee meetings and present the developments within the WGs.

The mission of the **Technological synergies WG** is to provide expertise in the field of nutrition contributing to knowledge-based development of novel food products, active ingredients, dietary patterns, and leveraging technologies to the field. Beyond the scope of the original proposal of each single project, the Technological synergies WG aims to develop new food products and technology services for personalized nutrition. **The WG possesses expertise in gut health, bioinformatics, machine learning, carbohydrate chemistry, *in-vitro* culture, and modelling of gut ecosystems. Additionally, it draws upon knowledge in plasma extracellular vesicles, glycomics, proteomics, microalgae strain modification, and advanced *in-vivo* monitoring of humans. Complementary innovations range from biointernet technologies and biosensor solutions to citizen science engagements. These capabilities are crucial for the WG's mission to develop and leverage cutting-edge nutrition technologies, aiming at solving health challenges through enhanced dietary patterns and optimized gut microbiome interactions.**

The mission of the **Regulatory activities and clinical trials synergies WG** is to support scientific advancement of the portfolio to ensure credible claims, consumer trust, and a forward-thinking regulatory framework across Europe. One portion of this mission requires navigating current regulations related to EFSA health and nutrition claims, EU novel foods, clinical trials, data protection (e.g. GDPR), and incorporating compliance with EU AI regulation. A larger effort of the working group is to address current emerging precision nutrition needs, such as implementation of novel biomarker-based clinical interventions and SOPs to harmonise precision nutrition clinical concepts. Furthermore, the group will carry out health/benefit risk analyses, going beyond general safety assessments for foods. Finally, the group will integrate ethical perspectives, such as the use of multi-omics approaches in consumer populations. The use of AI will be integrated into these ethical considerations.

The mission of the Business development WG is to support the exploitation of the results, and the emergence of a new generation of European companies at the forefront of personalised nutrition, capable of revolutionising health and well-being. This group aims to position the portfolio as an active player in the development of novel food systems that integrate cutting-edge knowledge of the gut microbiome, glycobiology, dietary interventions, and human systems biology.

By identifying and developing synergistic opportunities within and beyond the portfolio, the WG will strengthen inter-project connectivity, attract strategic partnerships, and increase the visibility of the portfolio in European and global innovation ecosystems, to inform political stakeholders and healthcare systems towards a prevention disease culture and to potentiate disease treatments through improved nutrition programs . Through targeted engagement with investors, industry stakeholders, and regulatory bodies, we will work to de-risk innovation pathways and create a robust foundation for business growth and scalability.

Key Objectives:

- Coordinate and align business development activities at the portfolio level.
- Identify and exploit synergies on business and scientific level between Pathfinder projects and with external partners, and assess the possibilities for follow-up funding within, and outside, the EIC programme.
- Evaluate market demand and customer personas regularly to anticipate future trends and adjust strategies accordingly.
- Support the creation of joint value propositions and commercial strategies.
- Increase visibility and positioning of the portfolio in relevant markets and sectors, as well as institutional relationships.
- Facilitate access to investment and industry partnerships.
- Support business readiness and go-to-market planning for emerging companies.
- Encourage knowledge exchange and best practice sharing in commercialisation strategies.
- Contribute to strategic dialogues with relevant EU and international stakeholders.

This WG will work in close collaboration with project leaders, EIC Programme Managers, the other WGs of the Portfolio and external advisors to ensure that the transformative science behind each project reaches its full commercial potential.

The mission of the Communication and Dissemination Working Group is to foster collaboration and alignment across the Portfolio, ensuring that the eight pioneering projects speak with one voice to maximise their collective impact. By identifying synergies and shared narratives, the WG aims to amplify the portfolio's visibility, increase outreach effectiveness,

and build a coherent and compelling presence across key audiences, including scientific communities, industry stakeholders, investors, policy makers, and the general public. Through strategic, inclusive, and forward-looking communication, this WG supports the portfolio's ambition to revolutionise health and well-being by advancing personalised nutrition strategies that leverage the complex relationships between diet, the gut microbiome, glycans, and human biology. This group will raise societal awareness about consumer behaviours and support the concepts of nutrition in food instead of supplements.

3. PORTFOLIO ACTIVITIES

The portfolio activities target three key dimensions reflected in the portfolio WGs: 1) **technological synergies**, 2) **regulatory synergies and clinical trials**, 3) **business development,** all of them connected and coordinated through the Steering Committee and the **communication and dissemination group**, established to support and facilitate the collaboration between the projects.

Technological synergies activities

Table 4 describes the technological synergies areas where commonalities among the projects have been identified, that can bring to specific technology related activities among projects.

Activity	Contributors	Description	Timeline
Develop a probiotic product for developing a urate-lowering microbial therapy	BUGS4URATE	• Develop a probiotic-based precision nutrition intervention • Identify and stratify responders vs. non-responders • Assess the mode-of-action and clinical efficacy of a novel probiotic and dietary fibre combination. • Study microbiome-host interaction in urate metabolism. • Translate findings into a food-grade product and develop commercialization pathways.	M18 – M40
Develop Microalgae strains and insect products for precision nutrition **from**	DIBAN	• Develop novel microalgae and insect products with high value biocompounds. • Identify innovative applications for microalgae and insects focused on human nutrition.	M18 – M40

microalgae and insects.		• Assess the effect of the developed microalgae on insect nutrition and composition. • Development of novel ex-vivo models for evaluation of microbiome-host interaction applied to novel food development using as starting model microalgae-fed insects. • Develop a personalized application based on continuous monitorization for precision nutrition. • Translate findings into a food-grade product and develop commercialization pathways.	
Identify biomarkers and dietary components that influence immune health, uncover how diet–microbiome interactions shape immunity, and provide a scientific basis for future nutrition guidelines supporting immune resilience.	NUTRIMMUNE	• Develop and validate composite immune biomarkers integrating multi-omic and clinical data to predict immune resilience and dietary response. • AI/ML models integrating multi-omics and lifestyle data to identify candidate immune biomarkers. • Predictive ingredient-level mapping, linking specific dietary molecules (fibres, polyphenols, immunonutrients) to immune biomarkers, guiding EU novel food formulations and clinical endpoints. • microbiome-derived metabolites (e.g., SCFAs, bile-acid derivatives) associated with host immune profiles; • Application programming interface (API) enabling industrial partners to perform predictive simulations on drug or ingredient candidates using a computational platform built on large patient-derived multi-omic and clinical datasets (metabolite predisposition modeling) • Clinical decision support module for clinicians to visualize biomarkers, assess risk, and generate nutrition recommendations, with interoperable APIs and regulatory documentation compliant with MDR. • Patient app delivering personalized meal plans, syncing with wearables, and tracking adherence/PROs; secure data exchange with HP Tool for closed-loop nutrition care.	M18 – M40

HMOs and L. paracasei synbiotics potential application	INTERACTHOMIGBUGS4URATE	• In silico analysis of L. paracasei for HMOs metabolism capability • Growth test to investigate the in vitro growth of L. paracasei using HMOs as carbon source • Investigating microbiota modulation effect and specificity of HMOs using Cominigut	M18 – M40

Table 4 : Technological synergies for portfolio activities.

Regulatory activities and clinical trials synergies

Table 5 describes the regulatory and clinical trials areas where the projects of the Precision Nutrition Portfolio could collaborate as well as potential activities.

Activity	Contributors	Description	Timeline
Harmonise regulatory and ethical aspects within multi-centre clinical trials	MENTORING, NUTRIMMUNE, GLUCOTYPES	Consider how different legal frameworks may drive the design and implementation of multi-centre clinical trials, also when samples are travelling within the EU for omics analyses.	M1 – M42
Individual health benefit assessment (PM) versus assessing evidence on group level (EBM)	ALL (lead by FIBRE-MATCH and Bugs4Urate)	Develop (clinical trial design/ bioinformatic) strategies that indicate potential individual health benefit rather than classical evidence-based medicine (benefit on group level).	M6 – M24
Exploratory feasibility study on regulatory aspects of the potential novel foods	DIBAN	Carry out a tentative, first-step assessment simulating the regulatory feasibility of the potential novel foods developed. This exploratory exercise will be based on the EFSA Novel Food Regulation, aiming to identify key considerations and anticipate requirements	M18 – M42

		that would be involved in preparing a formal technical dossier.	
Integration of nutritional intervention data for regulatory readiness	NUTRIMMUNE, MENTORING, FIBRE-MATCH	Assess regulatory aspects and data quality standards for nutrition-based clinical interventions. Explore how nutritional biomarkers and intervention outcomes can be integrated into regulatory evidence frameworks (e.g., EFSA health claims, clinical data reporting).	M12 – M30
Data integration and reusability, AI integration	ALL	Assess the regulatory implications/constraints for practical data reusability and data sharing; open data science.	M21 – M30

Table 5: Regulatory activities and clinical trials synergies portfolio activities.

Business development

Table 6 describes the business development areas where the projects of the Precision Nutrition Portfolio could collaborate as well as potential activities.

Activity	Contributors	Description	Timeline
Market surveys on acceptance of precision nutrition solutions	INTERACTHOMIG	Consumer surveys to understand willingness to adopt new solutions, willingness to change dietary patterns, etc.	M15 – M48
Stakeholders and Market awareness on novel food products & benefits	DIBAN	Joint presentation of novel food products/ benefits/testing/survey systems to relevant stakeholders and the nutrition market (KMZ, Biospain).	M4 – M10
Identifying potential market demand + commercialization pathways	GLUCOTYPES	wearable-integrated precision nutrition tools, particularly those based on real-time glucose variability analysis.	M11 – M20

Translating fibre–immune interaction data into market-ready guidelines	NUTRIMMUNE, FIBRE-MATCH	Jointly develop evidence-based dietary guidelines and recommendations linking fibre intake and immune resilience, targeting the food industry and health sector stakeholders.	M30 – M48
Establishment of a biomarker hub for data-driven nutrition trials	NUTRIMMUNE, MENTORING	Technologies and pipelines that can be transitioned into diagnostic services or biomarker-based nutrition platforms.	M30 – M48
Development of glycoproteomics innovations	GLUCOTYPES, MENTORING		M12 – M24
Joint value proposition design	GLUCOTYPES	Participation in joint value proposition design for personalized nutrition interventions that leverage novel biomarkers (e.g. glycan profiles, AI-derived signatures) as differentiators in consumer and clinical markets.	M18 – M26
Development of commercial digital products for professional nutrition assessment and personalized client-focused precision nutrition solutions	MENTORING	Development of advanced software, digital systems, and innovative computing models for precision nutrition product creation (including AI-driven nutrition consulting, biomarker analysis, and more)	M24 – M36
Development of approaches for assessing novel gut health and microbial activity	FIBRE-MATCH	Development of advanced in vivo monitoring for humans, in vitro culturing and modelling of the gut ecosystem, carbohydrate chemistry analytics and expertise.	M30 – M40

Table 6: Business development portfolio activities.

Communication and dissemination

Table 7 describes the communication areas where the projects of the Precision Nutrition Portfolio could collaborate as well as potential activities.

Activity	Contributors	Description	Timeline
Create a Portfolio page or section on each project website	ALL	All projects are encouraged to create a page or section to present the portfolio and the eight projects on their project website.	M1–M3
Follow/Reshare/React to the portfolio projects' social media posts & accounts	ALL	All projects are encouraged to follow, to tag (@mentions) and to react, share and comment the posts from the other projects. A table with the projects' social media links has been circulated to facilitate this.	M1–M48
Use a common hashtag: #EICPrecisionNutrition	ALL	When posting content relevant to the whole portfolio on social media, using the dedicated hashtag will increase the cohesion and visibility of the portfolio.	M1–M48
Share resources/milestones of the other projects on the website (e.g. via news section)	ALL	WG members will regularly share internally news about their project (infographics, videos, new publications, milestones reached, etc) that they believe are interesting for the other projects. Each project can choose to include that news in their own channels (website section dedicated to news and events or to the portfolio, newsletter, social media, etc)	M12–M48
Create a portfolio flyer/brochure	NUTRIEV: for the design ALL: for content	The working group will create a common flyer presenting the portfolio and its eight projects, which will be made available for all projects. Beneficiaries can print this flyer to distribute in the events they participate in on behalf of their project.	M13–M18

Organize a joint Precision Nutrition Portfolio conference as common (final) event	. ALL	Projects will join forces to organise a joint "Portfolio conference" to disseminate the results. Tentative date: summer 2028. An organising committee may be set up at a later stage. All projects are welcome to join, especially those who have planned a budget for the organisation of a final dissemination event.	M40–M48

Table 7: Communication and dissemination portfolio activities.

4. CROSS-PROJECT COLLABORATIONS

Table 8 describes the collaborations between the projects within the Portfolio and their common activities.

Cross-project collaboration (Strategy Plan).xlsx

Table 8: Summary of potential cross-project collaborations.

Further steps

The value of an EIC portfolio is greater than the sum of the individual projects it contains. **Figure 2** represents the eight projects belonging to the portfolio, emphasizing how it encompasses all individual, bilateral and multilateral product and technology developments that extend beyond the scope of initially contracted projects. Exploring possible synergies among proprietary technologies and competencies within the portfolio is essential for its success.

The strategic plan is regularly updated and defines the activities within the portfolio; however, it will be revised whenever necessary to reflect new opportunities or changing priorities during the annual meetings. Possible outcomes include new joint project proposals to EIC Pathfinder and Transition programmes, fast-tracking to the Accelerator, new licensing and trade agreements, joint ventures, and much more.

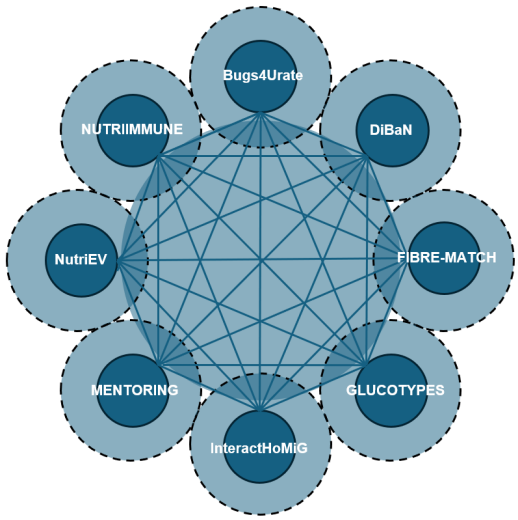


Figure 2. Schematic representation of collaboration among the eight projects within the Precision Nutrition portfolio.

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